

SYSTEM ARCHITECTURE BLUEPRINT

Loan Acceptance System

High Level Design Document for Automated Data-Driven Loan Processing &
Predictive Machine Learning Workflow

Executive Summary

Modernizing retail credit operations through automated risk assessment, scalable microservices, and high-precision machine learning.

| Background & Problem Statement

Industry Context

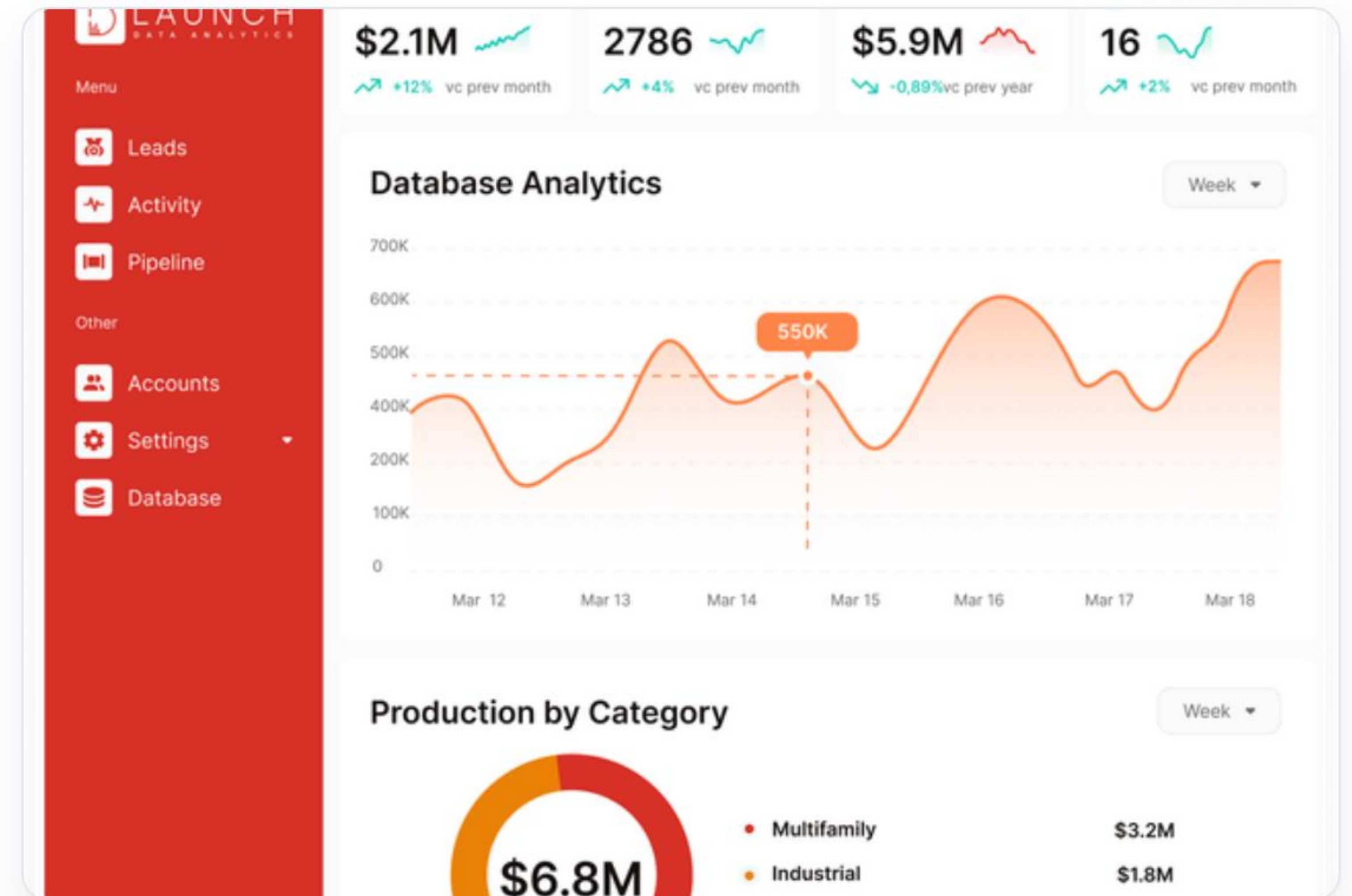
Loans are the primary profit driver for financial institutions, powered by interest income. Traditional verification processes are manual, slow, and often exclude creditworthy applicants who lack traditional banking histories.

Core Objective

Build an automated, electronic lending decision engine driven by ML algorithms. The system evaluates borrower parameters instantly to approve or reject applications with high predictive confidence.

| Data Collection Strategy

-  **Benchmark Sources:** Kaggle & UCI Repositories serve as baseline reference datasets for training initial model architectures.
-  **Credit Bureau Data:** Payment histories, credit scores, and past defaults collected from agency partners.
-  **Applicant Profiling:** Real-time demographic, employment, and income details captured securely via interactive UI.



| End-to-End System Workflow



1. Data Processing

Data Wrangling & Cleaning unifies complex multi-source inputs while Exploratory Data Analysis (EDA) identifies critical feature correlations.



2. Model Training

Feature selection feeds algorithms to train, evaluate, and tune models on holdout test splits to maximize prediction accuracy.

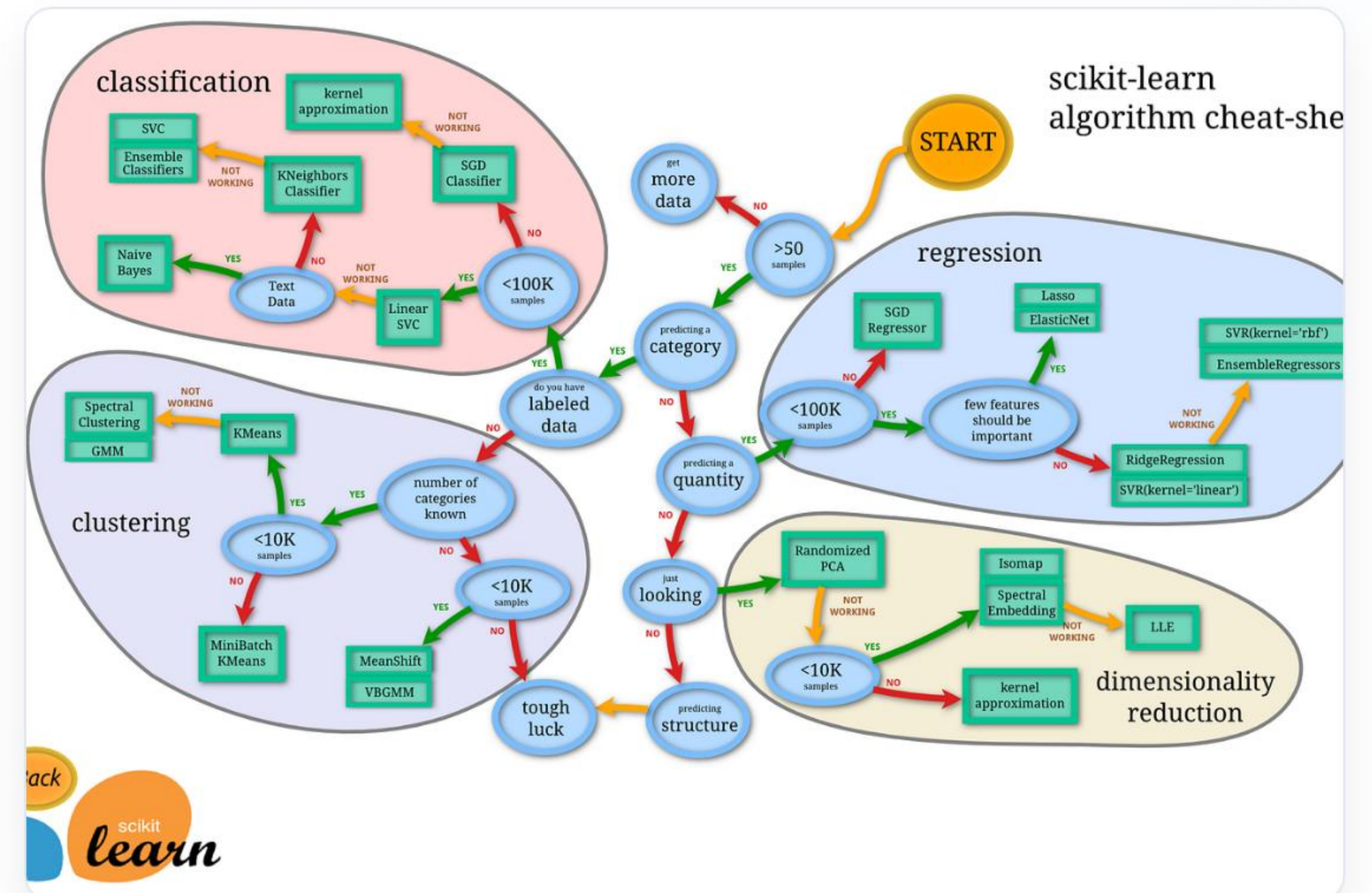


3. Production Deploy

Model packaging into Docker containers orchestrated via cloud infrastructure (GKE) for real-time API inference.

Machine Learning Engine

- Classification Logic:** Employs Decision Trees, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN).
- Ensemble Models:** Utilizes Random Forest (Bagging) and Gradient Boosting / AdaBoost to minimize variance.
- Automated Decisioning:** If calculated repayment probability is $\geq 75\%$, loan acceptance is auto-approved.



| Technology Stack Blueprint



Core Stack

Language: Python

Framework: Flask REST API

Database: MySQL Transactional Store



Frontend & ML

UI Layer: Streamlit Framework

ML Suite: Scikit-Learn Classification Models

Format: Pickle Serialization



DevOps & Cloud

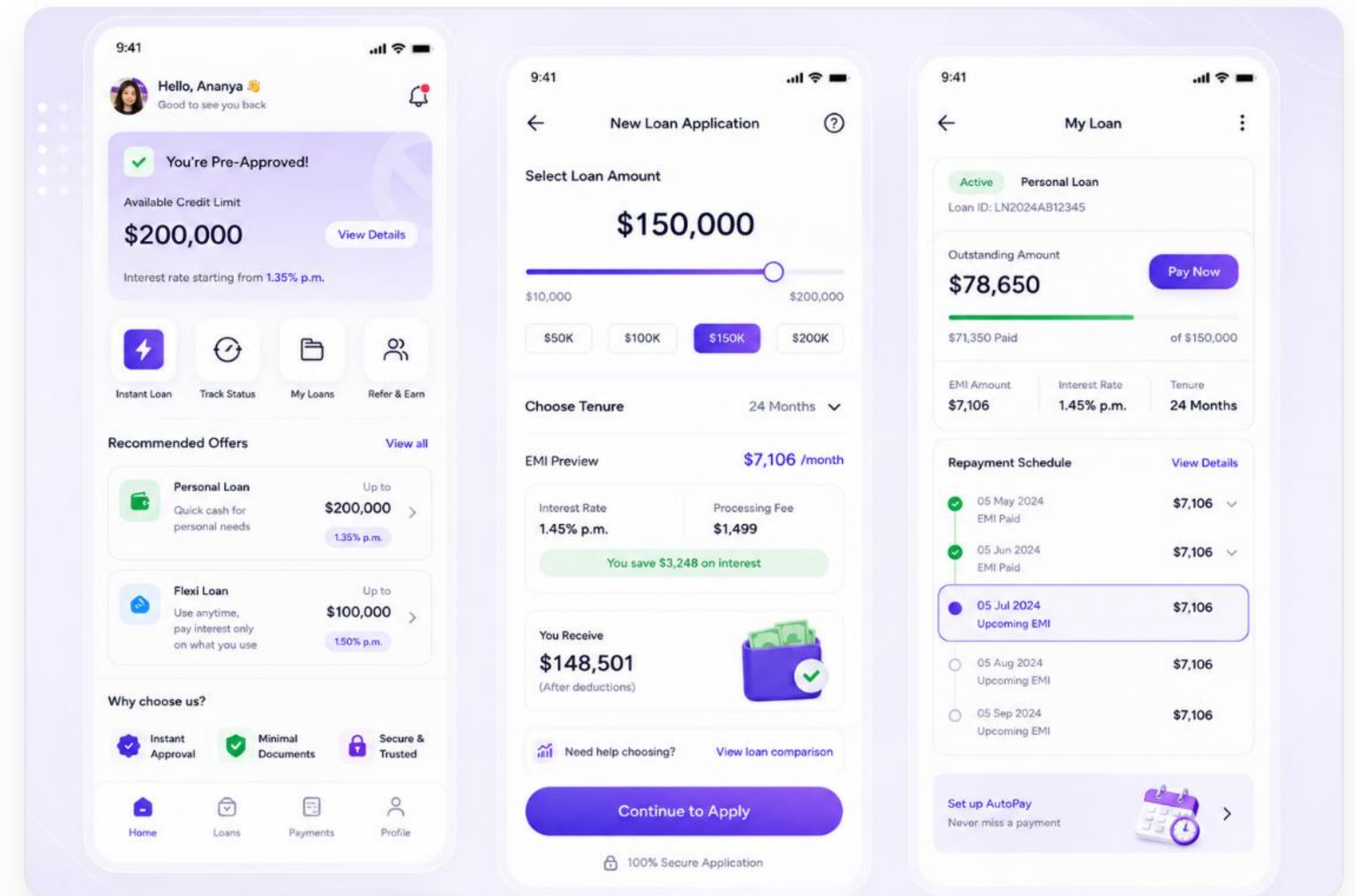
Container: Docker

Orchestrator: Kubernetes (GKE)

Pipeline: Automated CI/CD Workflows

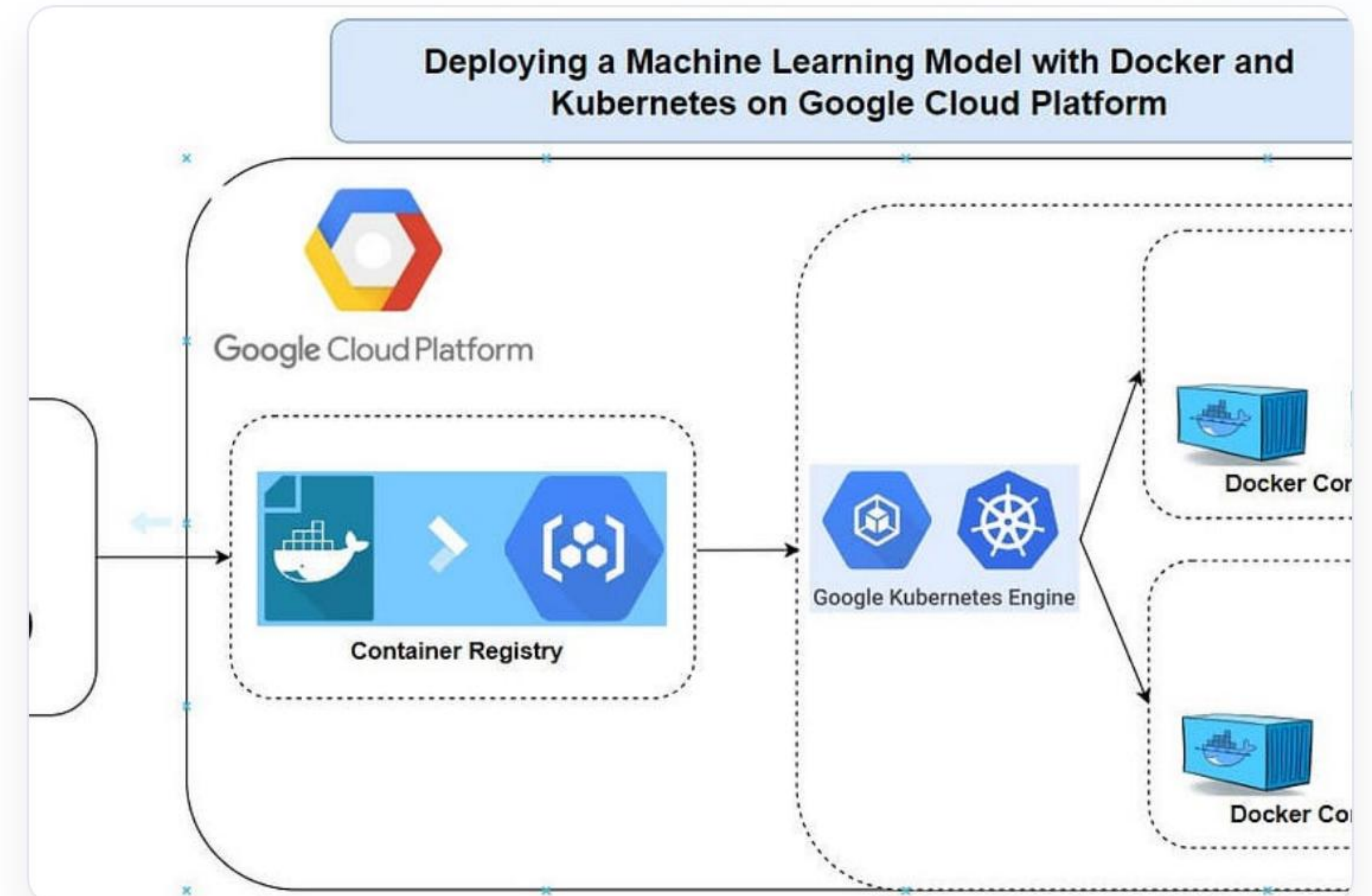
Web Application Architecture

-  **Front-End Interface (Streamlit):** Intuitive GUI supporting real-time single applications and batch CSV scoring.
-  **Back-End API (Flask):** Lightweight Python server hosting deserialized ML model pickles for API inference.
-  **Transaction Logging:** Stores decision outcomes and risk scoring in relational database logs.

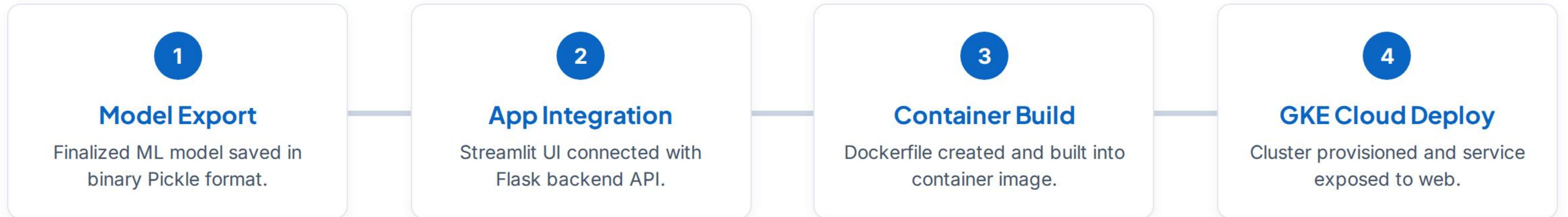


Deployment & Containerization

-  **Docker Containerization:** Standardized application packaging ensures environment parity between staging and production.
-  **Kubernetes Orchestration:** Managed GKE clusters handle automated autoscaling and load balancing.
-  **High Availability:** Zero-downtime rolling updates maintain continuous API availability.



| 4-Step Deployment Roadmap



| Business Impact & Speed

< 1s

Inference Execution Time

Operational Excellence

By transitioning from manual credit risk review to an automated ML decision engine, loan evaluation time is reduced from days to milliseconds.

The system expands loan access to underbanked demographics while drastically reducing default rates through objective model verification.

Image Sources



https://lirp.cdn-website.com/1aa69a8f/dms3rep/multi/opt/Mockup_for_Database_Analytics-640w.png

Source: www.launchcre.com



https://miro.medium.com/v2/resize:fit:1400/1*2NR51X0FDjLB13u4WdYc4g.png

Source: www.tzzminj.hr



<https://fenizomlmsoft.com/wp-content/uploads/2026/05/instant-loan-software-dashboard.png>

Source: fenizomlmsoft.com



https://miro.medium.com/v2/resize:fit:1200/1*hcwxSdzdVKQNKkBpGBs41g.jpeg

Source: medium.com